

Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems

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Outline

Forecasting Stability

Experimental Design

Empirical Results

Conclusions

Forecasting Stability

What is Forecast Stability?

Horizontal stability

- ▶ Measures how forecasts change across *different future time point* from the *same origin*.

Vertical stability

- ▶ Measures how forecasts change for the *same future time point* when the model is re-estimated at *different origins*.

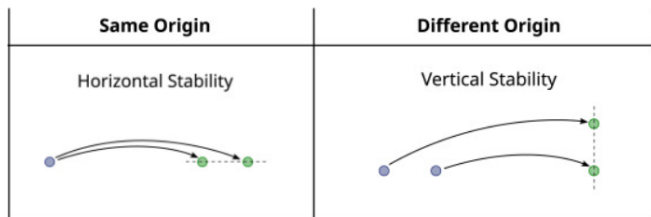


Figure 1: Godahewa, R., et al. (2025). *On forecast stability*.

Why it matters?

Forecasting drives operational decisions (inventory planning, workforce scheduling, budgeting and supply chain decisions, etc.)

Unstable forecasts lead to:

- ▶ revisions to plans and decisions,
- ▶ planning inefficiencies and higher costs,
- ▶ loss of credibility and trust in the forecasting system.

Accuracy–Stability Trade-off

Why can a trade-off arise?

- ▶ Forecasting models are updated frequently.
- ▶ More frequent retraining $\Rightarrow$ more accurate forecasts.
- ▶ Frequent model updates may produce structural model changes $\Rightarrow$ forecasts can shift.

Key questions

- ▶ Can we maintain high accuracy while producing stable forecasts?
- ▶ Is frequent retraining necessary?

Experimental Design

Datasets

A large forecasting competition dataset: M5 Competition.

Dataset	Frequency	N. Series	Test Set (T)	Horizon (h)
M5	Daily	28.298	364	28

Forecasting Models

Several different global forecasting models, from classical machine learning methods to deep learning architectures.

Machine Learning	Deep Learning
Linear Regression (LR)	MLP
XGBoost	TCN
LightGBM (LGBM)	NBEATSx
CatBoost	NHITS

Evaluation Strategy: Rolling Origin

- ▶ **Why?** Out-of-sample testing assesses generalization
- ▶ **How?**
 - ▶ Train/test split, preserving temporal order
 - ▶ Train model on initial training set
 - ▶ Forecast h steps ahead
 - ▶ Shift origin forward by p steps
 - ▶ Repeat until the test set is exhausted
- ▶ **Window types:**
 - ▶ Fixed window: Only the most recent n observations used
 - ▶ Expanding window: Include all historical data up to the current origin
- ▶ **In our study:** Expanding window with a step size $p = 1$

Evaluation Metrics

Accuracy

- ▶ **Point:** Root Mean Squared Scaled Error (RMSSE) (3)
- ▶ **Probabilistic:** Scaled Multi-Quantile Loss (SMQL) (4),(5)

Stability

- ▶ **Point:** Symmetric Mean Absolute Percentage Change (sMAPC) (6)
- ▶ **Probabilistic:** Symmetric Multi-Quantile Percentage Change (sMQPC) (1),(2)

Symmetric Multi-Quantile Percentage Change

The idea of sQPC is to mirror the sMAPC, but using quantile forecasts at different origins for some specific level α .

$$\text{sQPC}(\alpha) = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{q}_{t,n}^{(\alpha)} - \hat{q}_{t,n-1}^{(\alpha)}|}{|\hat{q}_{t,n}^{(\alpha)}| - |\hat{q}_{t,n-1}^{(\alpha)}|}. \quad (1)$$

The sMQPC simply averages the SQPC across the set of quantiles $\mathcal{Q}$, measuring overall probabilistic instability.

$$\text{sMQPC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \text{sQPC}(\alpha), \quad (2)$$

Sustainability Metrics

- ▶ Computing Time (CT)
- ▶ Cost (\$) and Savings (%)

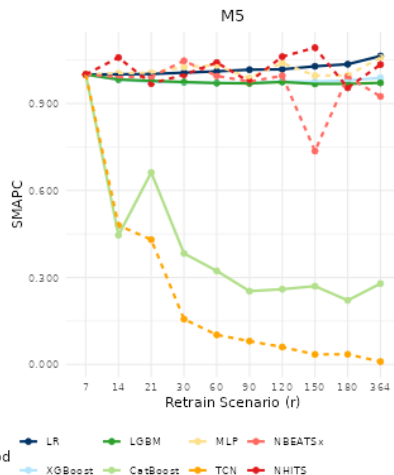
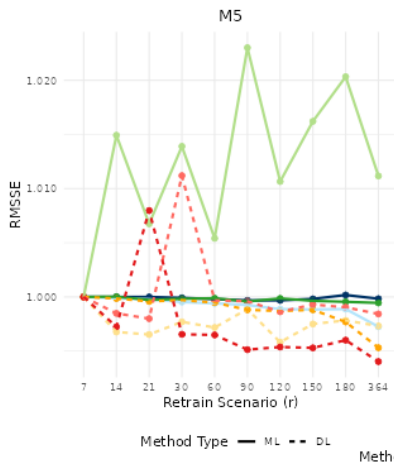
Simulation of the savings of each retraining scenario for a large retailer (200,000 SKUs and 5,000 stores), assuming some standard costs for computing services (\$3.5/hour).

Retraining Scenarios

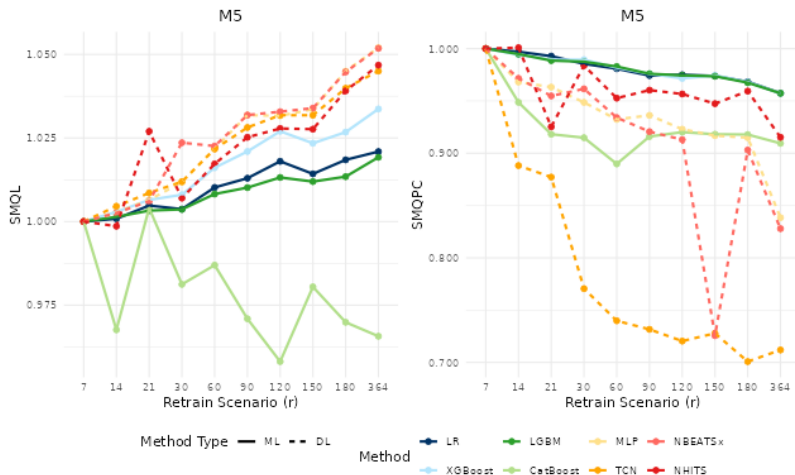
- ▶ **Retrain window (r):** the interval at which the model is re-trained or updated.
- ▶ **Scenarios based on data frequency:**
 - ▶ Daily: $r = \{7, 14, 21, 30, 60, 90, 120, 150, 180, 364\}$
- ▶ **Three strategies:**
 - ▶ Continuous retraining, $r = 7$, baseline scenario
 - ▶ No retraining, $r = T$
 - ▶ Periodic retraining, $7 < r < T$

Empirical Results

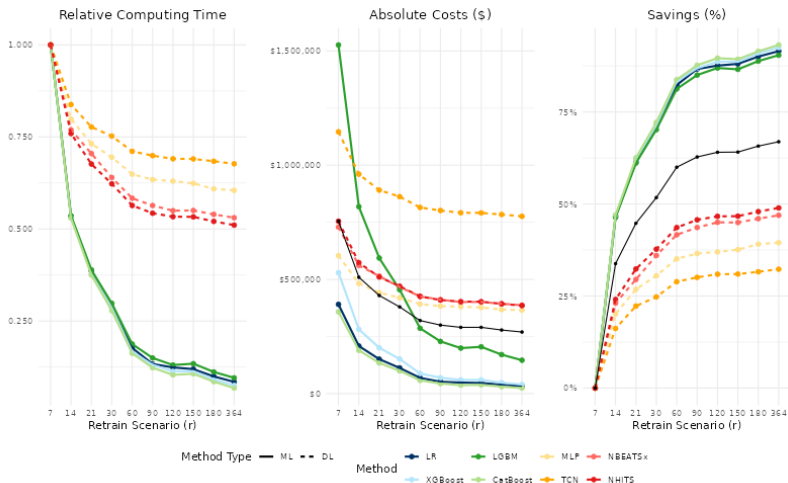
Point Forecasting



Probabilistic Forecasting



Sustainability



Conclusions

Key Results

Accuracy

- ▶ Point accuracy is stable across retraining scenarios
- ▶ Probabilistic accuracy slightly degrades with less frequent retraining

Stability

- ▶ Point stability is stable for lower retraining frequencies
- ▶ Probabilistic stability shows improvements for lower retraining frequencies

Sustainability

- ▶ Reducing the retraining frequency directly reduces the cost of producing forecasts up to 70% on average

Key Takeaways

- ▶ Highly frequent retraining may not be necessary
- ▶ Periodic retraining strategies allow for balancing accuracy and stability (especially in point forecasting)
- ▶ Reducing the retraining frequency supports more sustainable forecasting systems (lower costs of producing forecasts).

References

Zanotti, M. (2026). *Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems.*

arXiv. (url)

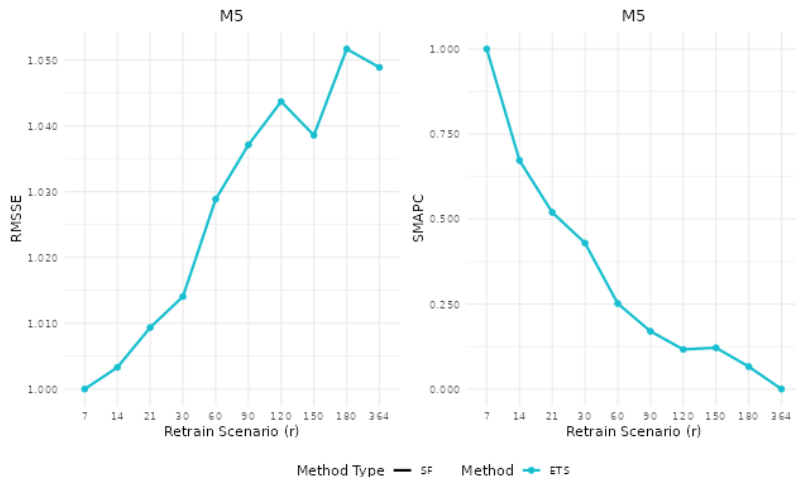
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Spiliotis, E., & Petropoulos, F. (2024). *On the update frequency of univariate forecasting models.* European Journal of Operational Research, 314, 111–121. (url)

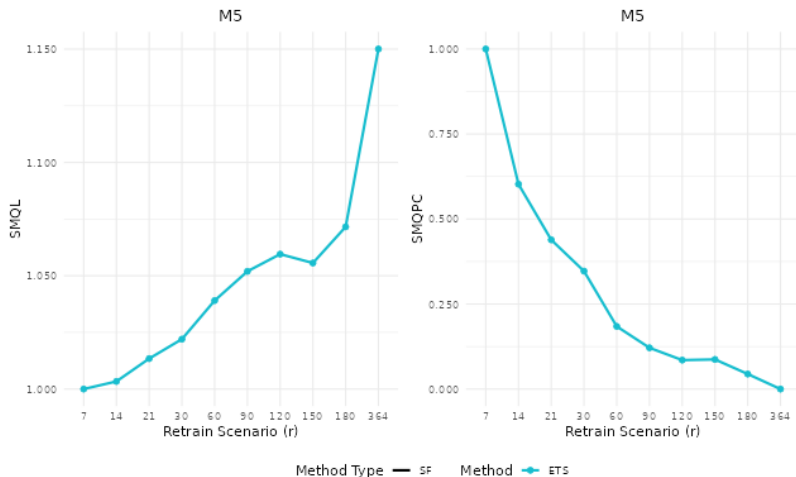
Thank you!

Appendix

Local Models - Point Forecasting



Local Models - Probabilistic Forecasting



Accuracy Metrics - Math

$$\text{RMSSE} = \sqrt{\frac{\frac{1}{h} \sum_{t=n+1}^{n+h} (y_t - \hat{y}_t)^2}{\frac{1}{n-s} \sum_{t=s+1}^n (y_t - y_{t-s})^2}} \quad (3)$$

$$\text{SQL}(\alpha) = \frac{\frac{1}{h} \sum_{t=n+1}^{n+h} \left(\alpha \cdot (y_t - \hat{q}_t^{(\alpha)}) \cdot \mathbb{I}_{\hat{q}_t^{(\alpha)} \leq y_t} + (1-\alpha) \cdot (\hat{q}_t^{(\alpha)} - y_t) \cdot \mathbb{I}_{\hat{q}_t^{(\alpha)} > y_t} \right)}{\frac{1}{n-s} \sum_{t=s+1}^n |y_t - y_{t-s}|} \quad (4)$$

$$\text{SMQL} = \frac{1}{|Q|} \sum_{\alpha \in Q} \text{SQL}(\alpha) \quad (5)$$

Stability Metrics - Math

$$\text{sMAPC} = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{y}_{t,n} - \hat{y}_{t,n-1}|}{|\hat{y}_{t,n}| - |\hat{y}_{t,n-1}|} \quad (6)$$

$$\text{sQPC}(\alpha) = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{q}_{t,n}^{(\alpha)} - \hat{q}_{t,n-1}^{(\alpha)}|}{|\hat{q}_{t,n}^{(\alpha)}| - |\hat{q}_{t,n-1}^{(\alpha)}|}. \quad (7)$$

$$\text{sMQPC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \text{sQPC}(\alpha), \quad (8)$$

Computing Setup

- ▶ Microsoft Azure NC6s v3 VM
- ▶ 6 vCPUs, 1 GPU, 112 GB RAM
- ▶ Nixtla's libraries: `statsforecast`, `mlforecast`, `neuralforecast`